

Project-Q Workbook - 4

30-Day Quantum Computing Challenge

DAY - 04

Superposition & Probability Amplitudes

Why Superposition Matters

Imagine you are searching for a friend in a city.

A classical computer follows one road at a time.

Road A



Road B



Road C



Answer

A quantum computer works differently.

Instead of committing to one possibility immediately, it can represent multiple possibilities within a quantum state.

This ability comes from a fundamental principle called:

Superposition

Without superposition:

No quantum computing.

No quantum algorithms.

No quantum advantage.

The Classical World

A classical bit can only be:

0 OR 1

Nothing in between.

Examples:

Light Switch:

ON = 1

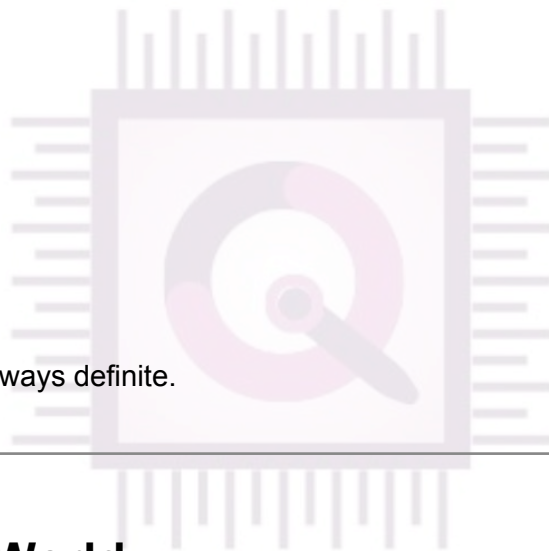
OFF = 0

Door:

Open = 1

Closed = 0

Classical information is always definite.



The Quantum World

A qubit behaves differently.

Before measurement:

$|0\rangle$

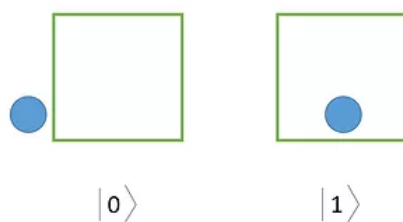
$|1\rangle$

or

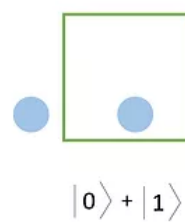
A Combination of Both

This combination is called: **Superposition**

Classical states



Quantum superposition state



What Is Superposition?

Superposition means: A qubit can exist in a quantum state that contains contributions from multiple basis states simultaneously.

For one qubit:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

This is called a superposition state.

Think About Music

Imagine pressing one piano key.

You hear: One Note

Now imagine pressing two keys together.

You hear: Multiple Notes

The sound now contains information from both notes.

Similarly,

A superposition state contains information from multiple basis states.

This analogy is closer to reality than saying:

"0 and 1 at the same time."

The Most Famous Superposition State

Consider:

$$|\psi\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$

This is called: **Equal Superposition**

Visual:

$|0\rangle$ 
 $|1\rangle$ 

Both possibilities contribute equally to the quantum state.

What Happens During Measurement?

Before Measurement:

$|\psi\rangle$

↓

Measurement

↓

$|0\rangle$ OR $|1\rangle$

After measurement, the superposition disappears.

This process is called: **State Collapse**

Coin Toss Analogy

A spinning coin is often used to visualize superposition.

While spinning:

Heads?

Tails?

Uncertain

After landing:

Heads OR Tails

Important:

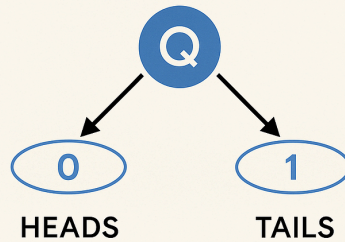
A spinning coin is **not actually a quantum superposition**.

It is only a visualization tool.

The quantum state is fundamentally different because it contains probability amplitudes and phase information.

Quantum Superposition

$$\alpha|0\rangle + \beta|1\rangle$$



Why Superposition Is Powerful

Suppose we have: 1 Qubit

Possible states:

$|0\rangle$

$|1\rangle$

Now: 2 Qubits

Possible basis states:

$|00\rangle$

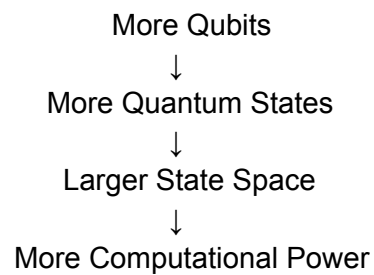
$|01\rangle$

$|10\rangle$

$|11\rangle$

A quantum system can exist in a superposition of all these basis states.

As qubits increase:



This is one reason quantum computing becomes interesting for complex problems.

Understanding Probability Amplitudes

Superposition introduces a new idea:

Probability Amplitudes

Most beginners think:

Probability and Amplitude are the same.

They are not.

Probability vs Probability Amplitude

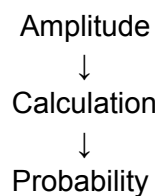
Probability tells us:

How likely an outcome is

Amplitude tells us:

Information used to calculate probability

Think of it like:



The Quantum State Formula

A qubit state is written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Where:

α = Probability Amplitude for $|0\rangle$

β = Probability Amplitude for $|1\rangle$

These amplitudes determine measurement outcomes.

Converting Amplitudes Into Probabilities

The rule is:

$$\text{Probability} = |\text{Amplitude}|^2$$

This rule is called: **Born Rule**

One of the most important rules in quantum mechanics.

Example 1

State:

$$|\psi\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$

Amplitude of $|0\rangle$: $1/\sqrt{2}$

Amplitude of $|1\rangle$: $1/\sqrt{2}$

Probability: $(1/\sqrt{2})^2 = 1/2 = 50\%$

Result:

$$P(|0\rangle) = 50\%$$

$$P(|1\rangle) = 50\%$$

Example 2

State:

$$|\psi\rangle = \sqrt{0.8} |0\rangle + \sqrt{0.2} |1\rangle$$

Probabilities:

$$P(|0\rangle) = 0.8 = 80\%$$

$$P(|1\rangle) = 0.2 = 20\%$$

Visual:

$|0\rangle$ 

$|1\rangle$ 

Measurement is more likely to produce: $|0\rangle$

Every valid quantum state must satisfy:

$$|\alpha|^2 + |\beta|^2 = 1$$

Why?

Because measurement must produce:

$|0\rangle$ OR $|1\rangle$

Total probability must always equal: 100%

A Hidden Feature: Phase

Probability amplitudes can be:

- Positive
- Negative
- Complex

This introduces: **Quantum Phase**

Example:

$$(|0\rangle + |1\rangle)/\sqrt{2} \text{ and } (|0\rangle - |1\rangle)/\sqrt{2}$$

Both give:

50%

50%

after measurement.

Yet they behave differently inside quantum algorithms.

This difference comes from phase.

Phase is what enables quantum interference.

Why Probability Amplitudes Matter

Quantum algorithms do not directly manipulate probabilities.

They manipulate:

Probability Amplitudes

The algorithm changes amplitudes so that:

Correct Answers: $\uparrow$ Amplified

Incorrect Answers: $\downarrow$ Suppressed

When measurement finally occurs:

The correct answer becomes more likely.

This idea powers many quantum algorithms.

Real-World Analogy

Imagine a voting competition.

Initially:

Team A = 50 Votes

Team B = 50 Votes

Campaign efforts increase votes for Team A.

After amplification:

Team A = 95 Votes

Team B = 5 Votes

Quantum algorithms perform a similar process on probability amplitudes before measurement.

Quick Comparison

Concept	Meaning
Superposition	Combination of multiple basis states
Probability Amplitude	Mathematical value inside a quantum state
Probability	Chance of measuring an outcome
Born Rule	Probability = $ \text{Amplitude} ^2$
Measurement	Converts quantum state into a classical outcome
Phase	Hidden property affecting interference

End-of-Day Summary

What You Learned Today

- ✓ What superposition really means
 - ✓ Why qubits differ from classical bits
 - ✓ Equal superposition states
 - ✓ State collapse during measurement
 - ✓ Probability amplitudes
 - ✓ Born Rule
 - ✓ Normalization condition
 - ✓ Difference between amplitudes and probabilities
 - ✓ Introduction to quantum phase
 - ✓ Why amplitudes matter in quantum algorithms
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Top Verified Learning Resources

IBM Quantum Learning (Highest Priority)

1. Qubits and Superposition

[IBM Quantum Learning Platform](#)

Study sections:

- Qubits
- Superposition
- Measurement
- Quantum States

These are IBM's official educational materials and align closely with modern quantum computing curricula.

2. Utility-Scale Quantum Computing Course

[IBM Quantum Course: Bits, Gates and Circuits](#)

Focus on:

- Quantum states
 - Superposition
 - Measurement probabilities
 - Bloch Sphere intuition
-

Microsoft Quantum Learning

[Microsoft Azure Quantum Documentation](#)

Particularly useful for understanding:

- Probability amplitudes
 - State vectors
 - Measurement outcomes
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Video Resources

1. Superposition Explained Clearly

[Qiskit Global Summer School YouTube Channel](#)

Search:

"Superposition and Quantum States"

2. Probability Amplitudes & Measurement

[Quantum Country](#)

One of the best beginner-friendly explanations of amplitudes and measurement.

